Example 7.11

Find the equations of tangent and normal to the curve $y = x^2 + 3x - 2$ at the point $(1, 2)$.

Solution

We have, $\frac{dy}{dx} = 2x + 3$. Hence at $(1, 2), \frac{dy}{dx} = 5$.

Therefore, the required equation of tangent is

$(y - 2) = 5(x - 1) \Rightarrow 5x - y - 3 = 0$.

The slope of the normal at the point $(1, 2)$ is $-\frac{1}{5}$.

Therefore, the required equation of normal is

$(y - 2) = -\frac{1}{5}(x - 1) \Rightarrow x + 5y - 11 = 0$.

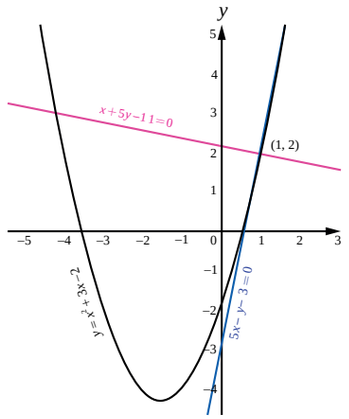


Fig.7.8

Example 7.12

Find the points on the curve $y = x^3 - 3x^2 + x - 2$ at which the tangent is parallel to the line $y = x$.

Solution

The slope of the line $y = x$ is 1. The tangent to the given curve will be parallel to the line, if the slope of the tangent to the curve at a point is also 1. Hence,

$$\frac{dy}{dx} = 3x^2 - 6x + 1 = 1$$

which gives $3x^2 - 6x = 0$.

Hence, $x = 0$ and $x = 2$.

Therefore, at (0, –2) and (2, –4) the tangent is parallel to the line $y = x$.

Example 7.13

Find the equation of the tangent and normal at any point to the Lissajous curve given by $x = 2\cos 3t$ and $y = 3\sin 2t, t \in \mathbb{R}$.

Solution

Observe that the given curve is neither a circle nor an ellipse. For your reference the curve is shown in Fig. 7.9.

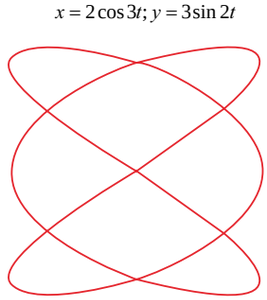
$$\text{Now, } \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$= -\frac{6\cos 2t}{6\sin 3t} = -\frac{\cos 2t}{\sin 3t}.$$

Therefore, the tangent at any point is

$$y - 3\sin 2t = -\frac{\cos 2t}{\sin 3t}(x - 2\cos 3t)$$

That is, $x\cos 2t + y\sin 3t = 3\sin 2t \sin 3t + 2\cos 2t \cos 3t$.



Lissajous curve
Fig.7.9